Higher-Order Eigenvalue Decomposition (HOEVD)

1 Introduction

Higher-Order Eigenvalue Decomposition (HOEVD) is a method for computing a Tucker decomposition of a tensor by extracting dominant subspaces along each mode. It can be viewed as a natural extension of matrix PCA or SVD to higher-order tensors.

Let

$$\mathcal{F} \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_d}$$

be a tensor of order d. The goal is to approximate $\mathcal{F}$ as

$$\mathcal{F} \approx \mathcal{G} \times_1 U_1 \times_2 U_2 \cdots \times_d U_d,$$

where

$$U_k \in \mathbb{R}^{n_k \times r_k}, \qquad U_k^T U_k = I_{r_k},$$

and

$$\mathcal{G} \in \mathbb{R}^{r_1 \times r_2 \times \cdots \times r_d}$$

is the core tensor.

2 Mode Unfoldings

For each mode $k \in \{1, \ldots, d\}$, define the mode-k unfolding

$$F_{(k)} \in \mathbb{R}^{n_k \times \prod_{j \neq k} n_j}.$$

This matrix rearranges the tensor so that mode k indexes rows and all other modes are flattened into columns.

3 Mode-wise Covariance Matrices

For each mode k, define the covariance matrix

$$C_k := F_{(k)} F_{(k)}^T \in \mathbb{R}^{n_k \times n_k}.$$

Each C_k is symmetric and positive semidefinite, and admits an eigendecomposition

$$C_k = Q_k \Lambda_k Q_k^T,$$

where

$$\Lambda_k = \mathrm{diag}(\lambda_1^{(k)}, \ldots, \lambda_{n_k}^{(k)}), \quad \lambda_1^{(k)} \geq \cdots \geq \lambda_{n_k}^{(k)} \geq 0.$$

We define the factor matrix

$$U_k := Q_k(:, 1 : r_k).$$

Equivalently, if

$$F_{(k)} = \widetilde{U}_k \Sigma_k \widetilde{V}_k^T$$

is the SVD, then

$$U_k = \widetilde{U}_k(:, 1 : r_k).$$

4 Core Tensor

Once the matrices $U_1, \ldots, U_d$ are computed, the core tensor is obtained by projection:

$$\mathcal{G} = \mathcal{F} \times_1 U_1^T \times_2 U_2^T \cdots \times_d U_d^T.$$

The resulting approximation is

$$\mathcal{F} \approx \mathcal{G} \times_1 U_1 \times_2 U_2 \cdots \times_d U_d.$$

5 Algorithm

Given $\mathcal{F}$ and target ranks $(r_1, \ldots, r_d)$, HOEVD proceeds as follows:

1. For each mode $k = 1, \ldots, d$, compute $F_{(k)}$.
2. Form
$$C_k = F_{(k)} F_{(k)}^T.$$
3. Compute the top r_k eigenvectors:
$$U_k = \mathrm{eig}_{r_k}(C_k).$$
4. Compute the core tensor:
$$\mathcal{G} = \mathcal{F} \times_1 U_1^T \times_2 U_2^T \cdots \times_d U_d^T.$$
5. Output
$$\mathcal{F} \approx \mathcal{G} \times_1 U_1 \times_2 U_2 \cdots \times_d U_d.$$

6 Symmetric Case

If $\mathcal{F} \in \mathbb{R}^{n \times \cdots \times n}$ is symmetric, we seek

$$\mathcal{F} \approx \mathcal{G} \times_1 U \times_2 U \cdots \times_d U,$$

with a single matrix $U \in \mathbb{R}^{n \times r}$.

Define the aggregated covariance matrix

$$C := \sum_{k=1}^{d} F_{(k)} F_{(k)}^T.$$

Compute its eigendecomposition

$$C = Q \Lambda Q^T, \quad U = Q(:, 1 : r).$$

Then define the core tensor

$$\mathcal{G} = \mathcal{F} \times_1 U^T \times_2 U^T \cdots \times_d U^T.$$

7 Remarks

HOEVD is a non-iterative method and is often used as an initialization for iterative tensor decomposition algorithms such as HOOI or ALS. It provides a computationally efficient way to extract dominant multilinear structure from high-dimensional data.